Develop Generative AI Applications: Get Started

Today's Skill Points
46 XP
See skill progress

Module 1
Foundations of Generative AI and Prompt Engineering

Welcome to the Course
Course Introduction
Video • 1 min
RAG and Agentic AI Professional Certificate Overview
Video • 5 min
Course Overview
Reading • 4 min
Helpful Tips for Course Completion
Reading • 5 min

(Optional) Generative AI Essentials
About this Optional Lesson
Reading • 2 min
Introduction to Generative AI
Video • 7 min
What are Generative AI Models?
Video • 8 min
Video: What is NLP (Natural Language Processing)?
Video • 9 min
Reading: Comprehensive Guide to Generative AI
Reading • 10 min
Practice Quiz: Generative AI Essentials
Practice Assignment • Grade: 100%

Working with Prompt Engineering and Prompt Templates
Introduction to In-Context Learning
Video • 5 min
Introduction to LangChain
Video • 3 min
Advanced Methods of Prompt Engineering
Video • 5 min
LangChain LCEL Chaining Method
Video • 4 min
(Optional) Reading: What is Prompt Engineering, and Why Do We Care?
Reading • 15 min
Choosing the Right Prompt Strategy: A Pre-Lab Readiness Check
Dialogue • 15 min
Lab: Master Prompt Engineering and LangChain PromptTemplates
Ungraded App Item • 1h
Practice Quiz: Working with Prompt Engineering and Prompt Templates
Practice Assignment • Grade: 100%

Module Summary and Evaluation
Summary and Highlights: Foundations of Generative AI and Prompt Engineering
Reading • 2 min
Cheat Sheet: Foundations of Generative AI and LangChain
Reading • 10 min
Graded Quiz: Foundations of Generative AI and Prompt Engineering
Graded Assignment • 21 min

Module 2
Introduction to LangChain in GenAI Applications

Module 3
Build a Generative AI Application with LangChain

Summary and Highlights: Foundations of Generative AI and Prompt Engineering

3 XP • Emerging ⊞ Practices & AI Integration

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Congratulations! You have completed this module. At this point in the course, you know:

- In-context learning is a prompt engineering method where demonstrations of the task are provided to the model as part of the prompt.
- Prompts are inputs given to an LLM to guide it toward performing a specific task.
- Prompt engineering is a process where you design and refine the prompt questions, commands, or statements to get relevant and accurate responses.
- Advantages of prompt engineering include that it boosts the effectiveness and accuracy of LLMs, ensures relevant responses, facilitates user expectations, and eliminates the need for continual fine-tuning.
- A prompt consists of four key elements: the instructions, the context, the input data, and the output indicator.
- Advanced methods for prompt engineering include zero-shot prompts, few-shot prompts, chain-of-thought prompting, and self-consistency.
- Prompt engineering tools can facilitate interactions with LLMs.
- LangChain uses 'prompt templates,' which are predefined recipes for generating effective prompts for LLMs.
- An agent is a key component in prompt applications that can perform complex tasks across various domains using different prompts.
- LCEL pattern structures workflows use the pipe operator (|) for clear data flow.
- Prompts are defined using templates with variables in curly braces {}.
- Components can be linked using RunnableSequence for sequential execution.
- RunnableParallel allows multiple components to run concurrently with the same input.
- LCEL provides a more concise syntax by replacing RunnableSequence with the pipe operator.
- Type coercion in LCEL automatically converts functions and dictionaries into compatible components.

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